



# Modelling Financial Markets

## Chapter 2: Market Efficiency — From EMH to Adaptive Markets



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## Acronyms used in this material (1/2)

- ▣ AIC: Akaike Information Criterion
- ▣ AMH: Adaptive Markets Hypothesis
- ▣ BDS: Brock–Dechert–Scheinkman (test)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BIST: Borsa İstanbul (BIST 100 index) (the Istanbul Stock Exchange (BIST 100 index))
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BUX: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CD: Chow–Denning (multiple variance-ratio test)
- ▣ CI: Confidence Interval
- ▣ CME: Chicago Mercantile Exchange
- ▣ CRSP: Center for Research in Security Prices
- ▣ DFA: Detrended Fluctuation Analysis
- ▣ EMH: Efficient Market Hypothesis
- ▣ ETF: Exchange-Traded Fund
- ▣ FRED: Federal Reserve Economic Data
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ FWER: Family-Wise Error Rate
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GMM: Generalized Method of Moments
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HK: Hong Kong
- ▣ IPC: Índice de Precios y Cotizaciones (the Mexican Stock Exchange index)
- ▣ IS: In-Sample

## Acronyms used in this material (2/2)

- ▣ MDH: Martingale Difference Hypothesis
- ▣ MIT: Massachusetts Institute of Technology
- ▣ ML: Machine Learning
- ▣ MSPE: Mean Squared Prediction Error
- ▣ NYSE: New York Stock Exchange
- ▣ OLS: Ordinary Least Squares
- ▣ OOS: Out-Of-Sample
- ▣ PX: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- ▣ RW1: Random Walk 1 (i.i.d. increments)
- ▣ RW2: Random Walk 2 (independent increments)
- ▣ RW3: Random Walk 3 (uncorrelated increments)
- ▣ SE: Standard Error
- ▣ SPA: Superior Predictive Ability (test)
- ▣ TOM: Turn Of the Month
- ▣ TSMOM: Time-Series MOMentum
- ▣ US: United States
- ▣ VR: Variance Ratio
- ▣ WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)



## Outline

Motivation

The Efficient Market Hypothesis

Testing Weak-Form Efficiency

Return Predictability from Valuation Ratios

Long Memory: Hurst, R/S and DFA

Adaptive Markets and Time-Varying Efficiency

Semi-Strong Efficiency: Event Studies

Anomalies

Behavioural Finance and Limits to Arbitrage

Efficiency and Machine Learning

Case Study: Market Efficiency in the Age of Big Data

AI for Scientific Discovery

Conclusions

References

## Can Anyone Beat the Market?

### The question of this chapter

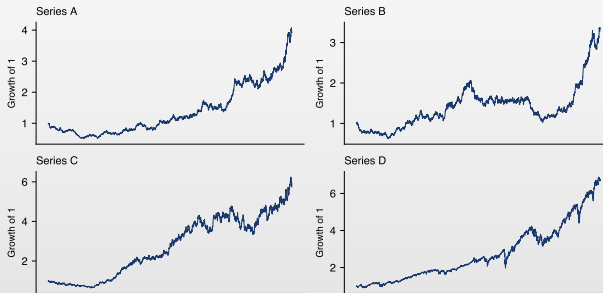
- If past prices revealed a strategy with positive expected risk-adjusted profit after costs, traders would exploit it until it disappeared
- A market in which this has already happened is called **efficient**
- The claim is testable: it restricts the statistical behaviour of returns

### What we do in this chapter

- Define efficiency precisely (Fama's three forms)
- Test it: autocorrelation, runs, variance ratios, long memory
- Ask whether efficiency changes over time (adaptive markets)
- Examine anomalies, behavioural finance and machine learning

## A Game: Which Series Is the Real Market?

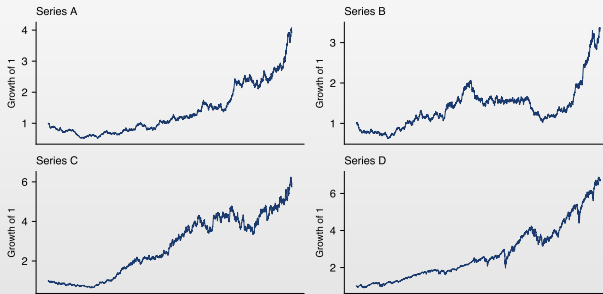
One of these is the S&P 500 (2010-2026); three are simulated random walks with the same drift and volatility



□ One panel is the S&P 500 (2010–2026); three are random walks with the same mean and volatility. Vote: A, B, C or D?

## A Game: Which Series Is the Real Market?

One of these is the S&P 500 (2010–2026); three are simulated random walks with the same drift and volatility



- One panel is the S&P 500 (2010–2026); three are random walks with the same mean and volatility. Vote: A, B, C or D?
- Answer: **D. Roberts (1959)**: random walks produce “patterns” by chance; we need statistics, not eyes

## Pioneers: Bachelier and Samuelson



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons



Paul A. Samuelson

Photo: Bernard Gotfryd (1970–1975); public domain; Wikimedia Commons

- Bachelier (1900): prices of French government bonds move like a random walk
  - ▶ “The mathematical expectation of the speculator is zero”
- Samuelson (1965): if prices are properly anticipated, their changes are unpredictable
  - ▶ Unpredictability is a sign of a well-functioning market, not of chaos
- Both results are the ancestors of the Efficient Market Hypothesis

## Data Used in This Chapter

| Group                    | Series                                                          | Period         |
|--------------------------|-----------------------------------------------------------------|----------------|
| Developed equity         | S&P 500, Euro Stoxx 50, Nikkei 225, Hang Seng                   | 2000–2026      |
| Emerging/frontier equity | BET, WIG20, BUX, PX, BIST 100, Bovespa, IPC, Nifty 50, Shanghai | 2000–2026      |
| Crypto-assets            | Bitcoin, Ethereum                                               | 2014/2016–2026 |
| Exchange rate            | EUR/RON, BNR reference rate                                     | 2005–2026      |

- Daily closing values up to 18 September 2026; returns are daily log returns  $r_t = \ln P_t - \ln P_{t-1}$
- Equity indices: weekdays only; a close equal to the previous close is treated as a holiday filled with the previous price and removed
  - ▶ The rule also drops the rare genuine zero-return day (S&P 500: 3 of about 6 700 days), which moves its test statistics by less than 0.05
- Crypto trades 7 days a week: 365 observations per year
- Each series is analysed on its own calendar; no series is joined to another in this chapter

## Eugene Fama and the Definition of Efficiency



Eugene F. Fama, Nobel Prize 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia

[Commons](#)

- **Fama (1970)**: a market is **efficient** if prices “fully reflect” all available information
  - ▶ Information set  $\mathcal{I}_t$ : everything known at time  $t$
- Formally: no strategy using  $\mathcal{I}_t$  earns returns above the fair (risk-adjusted) return
- Nobel Prize 2013, shared with Robert Shiller (sceptic, Section 9) and Lars Peter Hansen
- Update and survey: **Fama (1991)**; accessible critique: **Malkiel (2003)**

## Three Forms of Efficiency

| Form        | Information set $\mathcal{I}_t$          | Typical test                                              |
|-------------|------------------------------------------|-----------------------------------------------------------|
| Weak        | Past prices and returns                  | Autocorrelation, runs, variance ratios (Sections 3, 5, 6) |
| Semi-strong | All public information (news, reports)   | Event studies (Section 7)                                 |
| Strong      | Public and private (insider) information | Returns of insiders and fund managers                     |

- Each form contains the previous one: strong  $\Rightarrow$  semi-strong  $\Rightarrow$  weak
- This chapter focuses on the weak form; Chapter 3 tests risk-adjusted returns with factor models

## Efficiency as a Fair Game: the Martingale

- **Fair game** for excess returns:  $\mathbb{E}[r_{t+1} - \mu_t \mid \mathcal{I}_t] = 0$ , with  $\mu_t$  the required (equilibrium) return
  - ▶ With a constant required simple return  $m$ , the total-return price discounted at  $m$ ,  $P_t/(1+m)^t$ , is a martingale [Campbell, Lo & MacKinlay \(1997\)](#); discounting at the risk-free rate gives a martingale only under the risk-neutral measure
  - ▶  $P_t$  itself is one only if  $m = 0$ : a zero expected **simple** return, not a zero expected log return
- “ $\mathbb{E}[P_{t+1} \mid \mathcal{I}_t] = P_t$ ” and “ $\mathbb{E}[\ln(P_{t+1}/P_t) \mid \mathcal{I}_t] = \mu$ ” are different restrictions
  - ▶ They differ by discounting and by the Jensen term:  $\mathbb{E}[\ln(1+R)] \approx \mathbb{E}[R] - \text{Var}(R)/2$
- Consequence:  $\text{corr}(\varepsilon_{t+1}, g(\mathcal{I}_t)) = 0$  for any function  $g$  of the past,  $\varepsilon_{t+1} = r_{t+1} - \mu_t$ 
  - ▶ In particular all autocorrelations  $\rho_k = \text{corr}(r_t, r_{t-k})$ ,  $k \geq 1$ , are zero when  $\mu_t = \mu$
- The martingale says nothing about the variance: volatility may be predictable (Chapter 1) without contradicting efficiency

## Random Walk Hypotheses RW1, RW2, RW3

- Log price  $p_t = \ln P_t$  with  $p_t = \mu + p_{t-1} + \varepsilon_t$ ; the three versions differ in what they assume about  $\varepsilon_t$  [Campbell, Lo & MacKinlay \(1997\)](#)
- **RW1:**  $\varepsilon_t$  i.i.d. (the strongest; rules out volatility clustering)
  - ▶ Rejected by the stylised facts of Chapter 1
- **RW2:**  $\varepsilon_t$  independent but not identically distributed (variance may change)
- **RW3:**  $\varepsilon_t$  uncorrelated:  $\text{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$ , while  $\text{Cov}(\varepsilon_t^2, \varepsilon_{t-k}^2) \neq 0$  is allowed
  - ▶ The relevant hypothesis for real returns: tests must be robust to heteroskedasticity

## The Joint-Hypothesis Problem

- ▣ “Fair return” requires a model of expected returns (e.g. CAPM, Chapter 3)
- ▣ Every test of efficiency is a **joint** test of efficiency and of the asset-pricing model [Fama \(1991\)](#)
  - ▶ A rejection may mean an inefficient market **or** a wrong model of risk
- ▣ Example: a strategy earns 5% per year above the index; it may simply carry more risk than the index
- ▣ At daily horizons expected returns are tiny, so the choice of model matters little: weak-form tests are the cleanest
- ▣ Efficiency can never be “proved”, only not rejected by a given test on given data

## The Grossman–Stiglitz Paradox

- Gathering information is costly (analysts, data, computing)
- If prices reflected **all** information, nobody would be paid for collecting it [Grossman & Stiglitz \(1980\)](#)
  - ▶ Then nobody would collect it, and prices could not reflect it
- Resolution: an **equilibrium degree of inefficiency**, just large enough to reward informed traders for their costs
- Markets with cheap information and low costs (large, liquid) should be closer to efficiency
- A testable prediction for our comparison of developed, emerging and crypto markets (Section 3)

## What the EMH Does Not Say

- It does not say prices are “correct” at every moment, only that errors are not predictable
- It does not say nobody beats the market: some will, by luck; the question is whether it persists
- It does not forbid predictable volatility: RW3 allows volatility clustering
- It does not forbid predictable **expected** returns if they compensate time-varying risk
- It does not say bubbles are impossible; it says they are hard to time (Chapter 17)

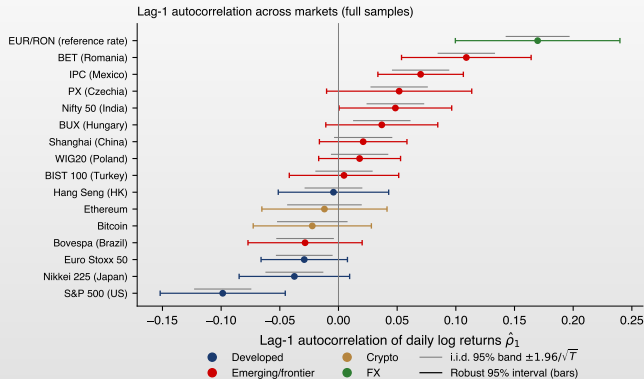
## Recap: the Efficient Market Hypothesis

- Efficient market: prices fully reflect the information set  $\mathcal{I}_t$  (weak, semi-strong, strong)
- Martingale / fair game: excess returns are unpredictable from  $\mathcal{I}_t$
- RW3 is the version to test; RW1 is already rejected
- Every test is joint with a risk model; costly information implies some inefficiency
- Small autocorrelations can be statistically significant and economically irrelevant

## Autocorrelation and Its Standard Error

- ▣ Sample autocorrelation:  $\hat{\rho}_k = \sum_{t=k+1}^T e_t e_{t-k} / \sum_{t=1}^T e_t^2$ , with  $e_t = r_t - \bar{r}$
- ▣ Under RW1 (i.i.d.):  $SE(\hat{\rho}_k) \approx 1/\sqrt{T}$ ; 95% band  $\pm 1.96/\sqrt{T}$ 
  - ▶ SE: standard error, the standard deviation of an estimator
- ▣ Unpredictable mean with volatility clustering (finite fourth moments):  $SE(\hat{\rho}_k) \approx \sqrt{\hat{\tau}_k}$ 
  - ▶  $\hat{\tau}_k = \sum_t e_t^2 e_{t-k}^2 / (\sum_t e_t^2)^2$  (robust variance)
  - ▶ When large moves cluster,  $\hat{\tau}_k > 1/T$ : the i.i.d. band is too narrow
- ▣ S&P 500: i.i.d. SE = 0.012, robust SE = 0.027, more than twice as large

## Lag-1 Autocorrelation Across Markets



□ Dots:  $\hat{\rho}_1$  with robust 95% intervals (bars); grey segments: the naive i.i.d. band

## Lag-1 Autocorrelation: Interpretation

- ▣ Positive and significant: BET 0.109 (robust SE 0.028), IPC 0.070, EUR/RON 0.170
  - ▶ Thin trading and slow diffusion of information (BET); a managed exchange rate (EUR/RON)
- ▣ Negative and significant: S&P 500  $-0.099$ 
  - ▶ Short-term overreaction and liquidity provision; concentrated in crises (Seminar B7)
- ▣ Bitcoin  $-0.022$  and Ethereum  $-0.012$ : not significant with robust errors
- ▣ Several i.i.d. “rejections” disappear with robust intervals: always use robust inference
- ▣ Economic size: regressing  $r_t$  on  $r_{t-1}$  gives  $R^2 = \hat{\rho}_1^2$ , i.e. 0.97% (S&P 500) and 1.19% (BET) of the daily variance

## Portmanteau Tests: Ljung–Box and Its Robust Version

- ▣ Ljung & Box (1978):  $Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k^2 / (T-k) \sim \chi_m^2$  under i.i.d. returns
- ▣ Robust version:  $\tilde{Q}(m) = \sum_{k=1}^m \hat{\rho}_k^2 / \hat{\tau}_k \rightarrow \chi_m^2$  under RW3 when the  $\hat{\rho}_k$  are asymptotically uncorrelated (condition on the  $z^*$  slide; Chapter 1)
- ▣  $m = 10$ , 5% critical value 18.31:
  - ▶ S&P 500:  $Q = 96.3$  but  $\tilde{Q} = 19.9$  ( $p = 0.031$ )
  - ▶ BET:  $Q = 108.0$ ,  $\tilde{Q} = 24.4$  ( $p = 0.007$ )
  - ▶ Bitcoin:  $Q = 20.2$  ( $p = 0.028$ ) but  $\tilde{Q} = 11.9$  ( $p = 0.294$ )
- ▣ The classical test rejects too often; the robust test retains the rejections for S&P 500 and BET, but not for Bitcoin

## The Runs Test

- Nonparametric: uses only the **signs** of returns (zero returns dropped), so it is immune to heavy tails [Wald & Wolfowitz \(1940\)](#)
- A **run** is a maximal sequence of equal signs: ++-+++ has 3 runs
- With  $n_1$  positive and  $n_2$  negative signs,  $n = n_1 + n_2$ , under independence:
  - ▶  $\mathbb{E}[R] = \frac{2n_1n_2}{n} + 1$ ,       $\text{Var}[R] = \frac{2n_1n_2(2n_1n_2 - n)}{n^2(n - 1)}$
  - ▶  $z = (R - \mathbb{E}[R]) / \sqrt{\text{Var}[R]} \approx N(0, 1)$
- Too **few** runs ( $z < 0$ ): signs persist (positive dependence of signs, not necessarily of returns); too **many** ( $z > 0$ ): sign reversals

## Runs Test on Real Markets

- ▣ S&P 500:  $z = 4.34$ : **more** runs than expected, consistent with  $\hat{\rho}_1 < 0$
- ▣ BET:  $z = -6.05$ : **fewer** runs, signs persist, consistent with  $\hat{\rho}_1 > 0$
- ▣ Nifty 50:  $z = -5.46$ ; IPC:  $z = -3.27$ : persistence in emerging markets
- ▣ Bitcoin  $z = 2.68$ , Ethereum  $z = 3.91$ : significant although  $\hat{\rho}_1$  is not
  - ▶ Signs and magnitudes carry different information: small reversals, large moves uncorrelated
- ▣ Caveat: the runs test assumes a constant probability of a positive day

## The Variance Ratio: Idea

- Uncorrelated increments with constant variance: variance grows linearly with the horizon,  
 $\text{Var}(r_t + \dots + r_{t-q+1}) = q \text{Var}(r_t)$ 
  - ▶  $q$ : the horizon in days (e.g.  $q = 5$  is one trading week)
  - ▶ With changing variances (RW2, RW3) the variance of the sum is the sum of the variances; the sample  $\widehat{\text{VR}}(q)$  still tends to 1 under the conditions of [Lo & MacKinlay \(1988\)](#)
- **Variance ratio:**  $\text{VR}(q) = \frac{\text{Var}(r_t + \dots + r_{t-q+1})}{q \text{Var}(r_t)}$ , equal to 1 for any covariance-stationary uncorrelated series
- For a covariance-stationary series, a weighted sum of autocorrelations:
  - ▶  $\text{VR}(q) = 1 + 2 \sum_{k=1}^{q-1} \left(1 - \frac{k}{q}\right) \rho_k$
- $\text{VR}(q) > 1$ : momentum (positive autocorrelation);  $\text{VR}(q) < 1$ : mean reversion
- Worked check: S&P 500  $1 + \hat{\rho}_1 = 0.901$  vs estimated  $\text{VR}(2) = 0.901$

## The Lo–MacKinlay Test

- Lo & MacKinlay (1988): overlapping  $q$ -day sums, with bias corrections

$$\blacktriangleright \hat{\sigma}_a^2 = \frac{1}{T-1} \sum_t (r_t - \hat{\mu})^2, \quad \hat{\sigma}_c^2(q) = \frac{1}{m} \sum_{t=q}^T \left( \sum_{j=0}^{q-1} r_{t-j} - q\hat{\mu} \right)^2$$

$$\blacktriangleright m = q(T - q + 1)(1 - q/T); \quad \widehat{VR}(q) = \hat{\sigma}_c^2(q) / \hat{\sigma}_a^2$$

- Under RW1 (homoskedastic):

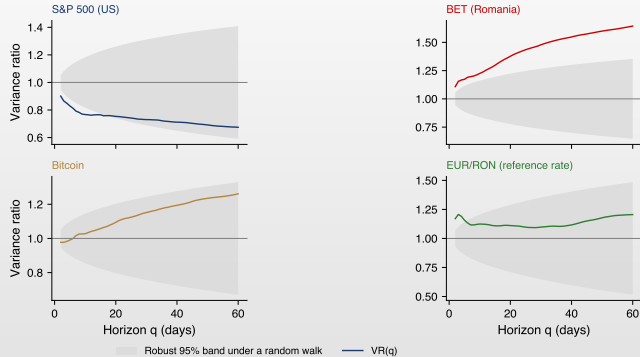
$$\blacktriangleright z(q) = \frac{\widehat{VR}(q) - 1}{\sqrt{2(2q-1)(q-1)/(3qT)}} \approx N(0, 1)$$

- Homoskedastic: constant variance; this  $z$  over-rejects when volatility clusters

## The Heteroskedasticity-Robust Statistic $z^*$

- Under RW3 plus the conditions of [Lo & MacKinlay \(1988\)](#) (finite moments, mixing,  $\mathbb{E}[\varepsilon_t^2 \varepsilon_{t-j} \varepsilon_{t-k}] = 0$  for  $j \neq k$ ), the  $\hat{\rho}_k$  are asymptotically uncorrelated with variances  $\hat{\delta}_k / T$ 
  - ▶  $\hat{\delta}_k = T \sum_{t=k+1}^T e_t^2 e_{t-k}^2 / \left( \sum_{t=1}^T e_t^2 \right)^2$
- Asymptotic variance of  $\widehat{VR}(q)$  and the test statistic:
  - ▶  $\hat{\theta}(q) = \sum_{k=1}^{q-1} \left( \frac{2(q-k)}{q} \right)^2 \hat{\delta}_k, \quad z^*(q) = \frac{\sqrt{T} (\widehat{VR}(q) - 1)}{\sqrt{\hat{\theta}(q)}} \approx N(0, 1)$
- $z^*$  tests RW3 under these conditions: they hold for GARCH with symmetric innovations, but can fail with a leverage effect
  - ▶ Without the last condition,  $\hat{\theta}(q)$  needs the full covariance matrix of the  $\hat{\rho}_k$
- S&P 500,  $q = 2$ :  $\widehat{VR} = 0.901$ ,  $z = -8.12$  but  $z^* = -3.64$ 
  - ▶ Still significant, but the homoskedastic statistic is more than twice as large in absolute value

## Variance-Ratio Profiles



□  $\widehat{VR}(q)$  for  $q = 2, \dots, 60$  days; grey: robust 95% band around 1 (from  $\hat{\theta}(q)$ )

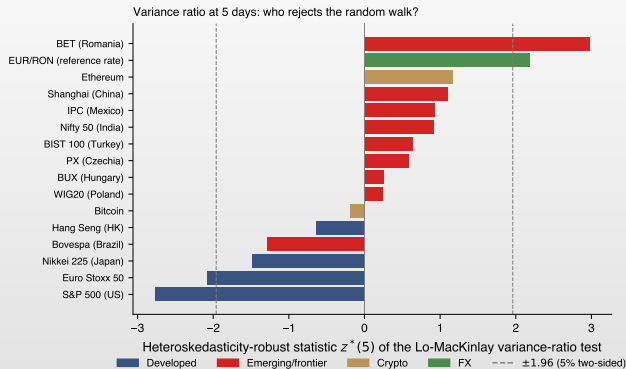
## Variance-Ratio Profiles: Interpretation

- ▣ S&P 500:  $VR = 0.901, 0.830, 0.753, 0.674$  at  $q = 2, 5, 20, 60$ : mean reversion at all horizons
- ▣ BET: 1.108, 1.173, 1.377, 1.644: persistence that accumulates with the horizon
- ▣ Bitcoin: 0.977, 0.991, 1.094, 1.261: inside the band up to about one month
- ▣ EUR/RON: 1.170, 1.157, 1.111: short-run persistence of a managed float
- ▣ The band widens with  $q$ : long horizons are informative about direction but tests there have low power

## Multiple Horizons: Chow–Denning and Choi

- ▣ Testing  $q = 2, 5, 10, 20$  separately inflates the probability of a false rejection
  - ▶ Four tests at 5% each: up to  $1 - 0.95^4 = 18.5\%$  if independent
- ▣ [Chow & Denning \(1993\)](#): use  $CD = \max_i |z^*(q_i)|$  and the studentised maximum modulus distribution
  - ▶ For  $m = 4$  horizons, the 5% critical value is 2.49 instead of 1.96
- ▣ [Choi \(1999\)](#): an automatic VR test that chooses the horizon from the data (spectral estimate at frequency zero)
- ▣ Wild-bootstrap versions for small samples: [Kim \(2006\)](#); applications to Asian markets: [Kim & Shamsuddin \(2008\)](#)

## Variance Ratio at One Week, Across Markets



Robust  $z^*(5)$  of the Lo-MacKinlay test for all 16 series; dashed lines:  $\pm 1.96$

## Variance Ratio Across Markets: Interpretation

- ▣ Reject at 5% with  $z^*(5)$ : BET (2.98), EUR/RON (2.19), S&P 500 ( $-2.77$ ), Euro Stoxx 50 ( $-2.08$ )
- ▣ Chow–Denning (horizons 2, 5, 10, 20) rejects for 5 of 16 series
  - ▶ BET  $p = 0.001$ , S&P 500  $p = 0.001$ , IPC  $p = 0.001$ , EUR/RON  $p = 0.000$ , Ethereum  $p = 0.027$
- ▣ Bitcoin:  $z^*(5) = -0.19$ , Chow–Denning  $p = 0.835$ : no evidence against RW3 over 2014–2026
- ▣ No simple ranking “developed = efficient, emerging = inefficient”: the direction differs (reversal vs persistence)

## Summary Table: Weak-Form Tests, 2000–2026

| Market                   | $T$   | $\hat{\rho}_1$ | robust SE | $\widehat{VR}(5)$ | $z^*(5)$ | CD $p$ | Runs $z$ |
|--------------------------|-------|----------------|-----------|-------------------|----------|--------|----------|
| S&P 500 (US)             | 6 714 | −0.099         | 0.027     | 0.830             | −2.77    | 0.001  | 4.34     |
| Euro Stoxx 50            | 6 836 | −0.029         | 0.019     | 0.910             | −2.08    | 0.097  | 3.41     |
| Nikkei 225 (Japan)       | 6 545 | −0.038         | 0.024     | 0.923             | −1.48    | 0.317  | 1.97     |
| Hang Seng (HK)           | 6 583 | −0.004         | 0.024     | 0.967             | −0.64    | 0.872  | −0.17    |
| BET (Romania)            | 6 670 | 0.109          | 0.028     | 1.173             | 2.98     | 0.001  | −6.05    |
| WIG20 (Poland)           | 6 685 | 0.018          | 0.018     | 1.009             | 0.24     | 0.777  | 2.25     |
| BUX (Hungary)            | 6 666 | 0.037          | 0.024     | 1.013             | 0.26     | 0.427  | −0.42    |
| PX (Czechia)             | 6 682 | 0.052          | 0.032     | 1.041             | 0.59     | 0.350  | −2.53    |
| BIST 100 (Turkey)        | 6 692 | 0.005          | 0.024     | 1.033             | 0.64     | 0.949  | −0.63    |
| Bovespa (Brazil)         | 6 622 | −0.028         | 0.025     | 0.928             | −1.29    | 0.521  | 0.88     |
| IPC (Mexico)             | 6 689 | 0.070          | 0.019     | 1.039             | 0.93     | 0.001  | −3.27    |
| Nifty 50 (India)         | 6 606 | 0.048          | 0.024     | 1.048             | 0.91     | 0.178  | −5.46    |
| Shanghai (China)         | 6 474 | 0.021          | 0.019     | 1.046             | 1.11     | 0.409  | 1.02     |
| Bitcoin                  | 4 384 | −0.022         | 0.026     | 0.991             | −0.19    | 0.835  | 2.68     |
| Ethereum                 | 3 913 | −0.012         | 0.027     | 1.067             | 1.17     | 0.027  | 3.91     |
| EUR/RON (reference rate) | 5 352 | 0.170          | 0.036     | 1.157             | 2.19     | 0.000  | −1.66    |

□  $T$ : number of daily returns; CD: Chow–Denning over  $q = 2, 5, 10, 20$ ; EUR/RON from July 2005, crypto on 365 days per year

## Beyond Linear Predictability: the Martingale Difference Hypothesis

- **MDH:**  $\mathbb{E}[r_t | \mathcal{I}_{t-1}] = \mu$ ; RW3 (all  $\rho_k = 0$ ) is weaker:  $r_t = \varepsilon_t + \theta(\varepsilon_{t-1}^2 - \sigma^2)$  has  $\rho_k = 0$  ( $\varepsilon_t$  i.i.d. symmetric) but a predictable mean
  - ▶ Generalised spectral test of the MDH, with wild-bootstrap  $p$ -values: [Escanciano & Velasco \(2006\)](#)
- [Escanciano & Lobato \(2009\)](#): automatic portmanteau  $AQ = T \sum_{j \leq \tilde{p}} \hat{\gamma}_j^2 / \hat{\kappa}_j \rightarrow \chi_1^2$ ,  $\hat{\gamma}_j = \frac{1}{T-j} \sum_t e_t e_{t-j}$ ,  $\hat{\kappa}_j = \frac{1}{T-j} \sum_t e_t^2 e_{t-j}^2$  (robust to conditional heteroskedasticity);  $\tilde{p} \leq 10$  maximises  $Q_p$  minus a penalty that switches from BIC ( $p \ln T$ ) to AIC ( $2p$ ) when some  $\sqrt{T}|\hat{\rho}_j|$  is large
  - ▶ AQ: S&P 500 13.1 ( $p = 0.0003$ ), BET 14.9 ( $p = 0.0001$ ), Bitcoin 0.8 ( $p = 0.385$ ); 5 of 16 series reject at 5%
- **BDS** [Brock, Dechert, Scheinkman & LeBaron \(1996\)](#) tests i.i.d., not the MDH: GARCH alone makes it reject
  - ▶ A BDS rejection is evidence against RW1, not against weak-form efficiency (RW3)

## Recap: Testing Weak-Form Efficiency

- ▣ Autocorrelations, Ljung–Box and variance ratios test linear serial dependence; runs test sign ordering; the MDH also rules out nonlinear predictability of the mean
- ▣ Volatility clustering makes classical standard errors too small: use  $\hat{\tau}_k$ ,  $\tilde{Q}$  and  $z^*$
- ▣ Several horizons need a joint test (Chow–Denning); several markets need care in reading “significance”
- ▣ S&P 500 shows reversal, BET and EUR/RON persistence; Bitcoin is not rejected on the full sample by the robust autocorrelation and VR tests
- ▣ Statistical rejection is not yet a profitable strategy: trading costs and risk decide

## Predictive Regressions and the Stambaugh Bias

- Model:  $r_{t+1} = \alpha + \beta x_t + u_{t+1}$ ,  $x_{t+1} = \theta + \rho x_t + v_{t+1}$ , with  $\text{Cov}(u, v) = \sigma_{uv}$ 
  - ▶  $r$ : excess return;  $x$ : log dividend-price ratio  $dp_t = \ln D_t - \ln P_t$ , persistent ( $\rho \approx 1$ )
- $x_t$  is predetermined, not strictly exogenous: OLS is consistent but biased [Stambaugh \(1999\)](#)
  - ▶ Project  $u$  on  $v$ :  $u = (\sigma_{uv}/\sigma_v^2)v + e$ , so  $\hat{\beta} - \beta = (\sigma_{uv}/\sigma_v^2)(\hat{\rho} - \rho) + \text{noise}$
  - ▶ With  $\mathbb{E}[\hat{\rho} - \rho] \approx -(1 + 3\rho)/T$ :  $\mathbb{E}[\hat{\beta} - \beta] \approx -\frac{\sigma_{uv}}{\sigma_v^2} \frac{1 + 3\rho}{T}$
- Price up  $\Rightarrow dp$  down:  $\sigma_{uv} < 0$ , so the bias is **upward**: predictability looks stronger than it is
- Near a unit root the  $t$ -statistic is not  $N(0, 1)$  either: valid inference by a Bonferroni  $Q$ -test over a confidence interval for  $\rho$  [Campbell & Yogo \(2006\)](#)

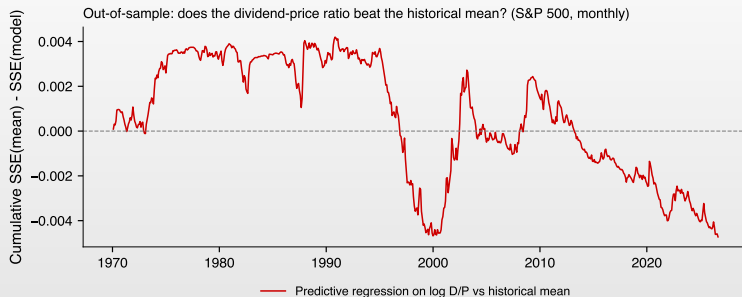
## The Stambaugh Bias in Numbers: S&P 500, 1934–2026

- Monthly log excess return (dividends included, minus the 3-month T-bill rate) on lagged  $dp_t$ ,  $T = 1\,111$  months
  - ▶ Data: [Shiller \(2026\)](#) (S&P Composite, to 2026-06), extended with our S&P 500 closes and ETF dividends; T-bill: FRED TB3MS
- OLS:  $\hat{\beta} = 0.267$  (% per month per unit of  $dp$ ), Newey–West  $t = 0.91$  (12 lags),  $R^2 = 0.12\%$
- Predictor:  $\hat{\rho} = 0.9988$ ; innovations:  $\text{corr}(\hat{u}, \hat{v}) = -0.97$ ,  $\hat{\phi} = \hat{\sigma}_{uv} / \hat{\sigma}_v^2 = -0.925$  (decimal units)
- Estimated bias  $-100 \hat{\phi}(1 + 3\hat{\rho}) / T = 0.333$  (in %, like  $\hat{\beta}$ ) exceeds  $\hat{\beta}$  itself: bias-corrected slope  $-0.066$ 
  - ▶ Corrected slope / unadjusted HAC SE =  $-0.22$ : only illustrative, it ignores the uncertainty of the correction; formal inference: [Campbell & Yogo \(2006\)](#) or a bootstrap
- Simulation with the same  $\rho$  and  $\text{corr}(u, v)$ ,  $\beta = 0$ ,  $T = 360$  months: the nominal 5%  $t$ -test rejects in 26.5% of samples

## Out-of-Sample Predictability: $R_{OS}^2$ and the Clark–West Test

- Recursive forecasts  $\hat{r}_t$  (expanding window) vs the historical mean  $\bar{r}_t$ :  $R_{OS}^2 = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t (r_t - \bar{r}_t)^2$ 
  - ▶ Most predictors have  $R_{OS}^2 < 0$  [Welch & Goyal \(2008\)](#); the 2022 update confirms it [Goyal, Welch & Zafirov \(2024\)](#)
  - ▶ Pseudo out-of-sample, as in [Welch & Goyal \(2008\)](#): Shiller's monthly dividends are interpolated from quarterly totals, so  $dp_t$  is not strictly real-time
- Nested models: under  $H_0$  the larger model estimates a zero  $\beta$ , which inflates its MSPE; Diebold–Mariano is undersized
  - ▶ [Clark & West \(2007\)](#):  $f_t = (r_t - \bar{r}_t)^2 - [(r_t - \hat{r}_t)^2 - (\bar{r}_t - \hat{r}_t)^2]$ ; one-sided  $t$  on the mean of  $f_t$
- [Campbell & Thompson \(2008\)](#): impose the sign of  $\beta$  and a non-negative forecast
  - ▶ Mean–variance timer: expected return rises by the proportion  $\frac{R^2}{1 - R^2} \cdot \frac{1 + S^2}{S^2}$ ,  $S$  the monthly Sharpe ratio

## Out-of-Sample Evidence: the Dividend-Price Ratio



- Cumulative squared-error gain over the historical mean, forecasts from 1970; rising line: the predictor helps

## Out-of-Sample Evidence: Interpretation

- ▣ 680 monthly forecasts from 1970:  $R_{OS}^2 = -0.51\%$ ; the Campbell–Thompson restrictions bind in 5.7% of months and give  $-0.51\%$
- ▣ Clark–West  $t = 0.42$  (one-sided  $p = 0.34$ ): no out-of-sample predictability from  $dp$
- ▣ The gain peaks in 1990 and is lost in the 1990s boom, when  $dp$  fell to record lows; after 2010 it erodes again
- ▣ Economic scale: with monthly  $S = 0.160$ , an  $R^2$  of only 0.5% would raise a timer's expected return by 20%
- ▣ Even a genuine slope would not reject the EMH: time-varying expected returns can be compensation for risk (joint hypothesis)

## Recap: Predictability from Valuation Ratios

- ▣ Persistent predictors with innovations correlated with returns bias  $\hat{\beta}$  upward and distort the  $t$ -test
- ▣ S&P 500 since 1934: the in-sample slope is insignificant and smaller than its own Stambaugh bias
- ▣ Out of sample, compare with the historical mean using  $R_{OS}^2$  and Clark–West, not Diebold–Mariano
- ▣ Small  $R^2$  can still be economically large: judge predictability by utility, not by  $R^2$  alone

## Harold Hurst and the Nile



Harold Edwin Hurst (1880–1978)

Photo: Elliott & Fry (1953); public domain; Wikimedia

[Commons](#)

- Hurst (1951) designed Nile reservoirs: wet years followed wet years more often than chance allows, measured with the rescaled range



Nilometer on Elephantine Island, Aswan

Photo: Olaf Tausch (2011); CC BY 3.0; Wikimedia Commons

## The Rescaled Range (R/S) Statistic

- For a window of  $n$  returns with mean  $\bar{r}_n$  and standard deviation  $s_n$ :
  - ▶ Cumulative deviations  $Y_k = \sum_{i=1}^k (r_i - \bar{r}_n)$ ,  $k = 1, \dots, n$
  - ▶ Range  $R_n = \max_k Y_k - \min_k Y_k$ ; rescaled range  $R_n/s_n$
- Scaling law:  $\mathbb{E}[R_n/s_n] \approx c n^H$ , where  $H$  is the **Hurst exponent**
  - ▶ Estimate: slope of  $\ln(R/S)$  on  $\ln n$ , averaging  $R/S$  over non-overlapping windows
- $H = 0.5$ : no long memory (short-run dependence allowed);  $H > 0.5$ : persistence (long memory);  $H < 0.5$ : anti-persistence
  - ▶ **Long memory**: autocorrelations decay so slowly that their sum diverges,  $\rho_k \sim k^{2H-2}$

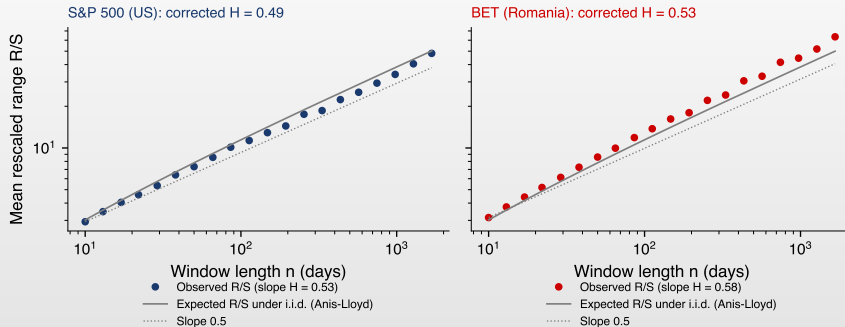
## Hurst Exponent and Variance Ratio

- ▣ For self-similar increments (fractional Gaussian noise):  $\text{Var}(r_t + \dots + r_{t-q+1}) = q^{2H} \text{Var}(r_t)$
- ▣ Hence  $\text{VR}(q) = q^{2H-1}$  and  $H = 0.5 + \ln \text{VR}(q) / (2 \ln q)$
- ▣ Worked example, Bitcoin:  $\widehat{\text{VR}}(20) = 1.094$ 
  - ▶  $\hat{H} = 0.5 + 0.0900/5.991 = 0.515$ , close to 0.5
- ▣ Same relation for BET ( $\widehat{\text{VR}}(20) = 1.377$ ) is solved in Seminar A5
- ▣ The relation holds only for exact self-similarity: short-memory processes give  $H(q)$  that depends on  $q$

## The Small-Sample Bias of R/S

- ▣ Even for i.i.d. returns, the slope of  $\ln(R/S)$  on  $\ln n$  exceeds 0.5 for small  $n$
- ▣ Anis & Lloyd (1976): exact  $\mathbb{E}[R_n/s_n]$  under i.i.d. Gaussian returns
  - ▶  $\mathbb{E}[R_n/s_n] = \frac{\Gamma((n-1)/2)}{\sqrt{\pi} \Gamma(n/2)} \sum_{i=1}^{n-1} \sqrt{\frac{n-i}{i}}$  ( $\Gamma$ : the gamma function)
- ▣ Corrected exponent:  $H_{AL} = 0.5 + \text{slope of } \ln(R/S) - \ln \mathbb{E}[R_n/s_n] \text{ on } \ln n$
- ▣ Uncorrected estimates overstate memory by about 0.04–0.05 with daily data (next chart)

## R/S Analysis: S&amp;P 500 and BET



□ Dots: average  $R/S$  for window lengths  $n$ ; grey line: the Anis-Lloyd expectation under i.i.d. returns

## R/S Analysis: Interpretation

- ▣ S&P 500: raw  $\hat{H} = 0.530$ , corrected  $\hat{H}_{AL} = 0.486$ : no long memory, slight anti-persistence
- ▣ BET: raw  $\hat{H} = 0.578$ , corrected  $\hat{H}_{AL} = 0.533$ : persistence remains after the correction
- ▣ Without the correction, the S&P 500 would look “persistent” ( $0.530 > 0.5$ ): a pure small-sample artefact
- ▣ Open question: is BET's  $H > 0.5$  **long** memory or just the short-run autocorrelation  $\hat{\rho}_1 = 0.109$ ?
  - ▶ Classical R/S cannot tell them apart: next slide

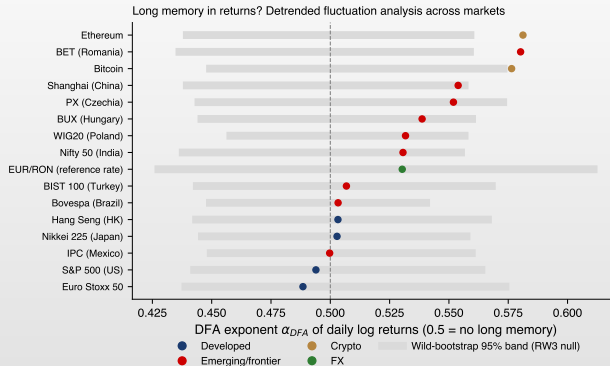
## Lo's Modified R/S Test

- Lo (1991): short-run autocorrelation inflates R/S and mimics long memory
- Replace  $s_n$  by a long-run (HAC) standard deviation with  $\ell$  lags:
  - ▶  $\hat{\sigma}_\ell^2 = \hat{\gamma}_0 + 2 \sum_{j=1}^{\ell} \left(1 - \frac{j}{\ell+1}\right) \hat{\gamma}_j$ ,  $\hat{\gamma}_j$ : autocovariance at lag  $j$
  - ▶  $V = \frac{\max_k Y_k - \min_k Y_k}{\hat{\sigma}_\ell \sqrt{T}}$ ; bandwidth  $\ell$  chosen as in Andrews (1991)
- Under short memory,  $V$  lies in  $[0.809, 1.862]$  with probability 95%
- Full samples: S&P 500  $V = 1.559$ , Bitcoin 1.547, PX 1.750, EUR/RON 1.429: not rejected
  - ▶ BET  $V = 1.961$  ( $\ell = 7$ ) and Ethereum  $V = 1.980$ : just above 1.862, weak evidence of long memory

## Detrended Fluctuation Analysis (DFA)

- Peng et al. (1994): robust to slow trends and non-stationary means
- Algorithm:
  - ▶ Profile  $y_k = \sum_{i=1}^k (r_i - \bar{r})$ ; split into boxes of length  $n$
  - ▶ In each box fit a linear trend and keep the residuals
  - ▶  $F(n)$  = root mean square of all residuals; repeat for many  $n$
- $F(n) \propto n^{\alpha_{DFA}}$ : the DFA exponent  $\alpha_{DFA}$  plays the role of  $H$  ( $\alpha_{DFA} = 0.5$ : no long memory)
- Null band by **wild bootstrap**:  $e_t \eta_t$ ,  $\eta_t = \pm 1$  with probability 1/2 (199 replications)
  - ▶ Keeps the volatility path  $|e_t|$  (allowed under RW3) and destroys serial correlation; shuffling would destroy both

## DFA Exponent Across Markets



□ Dots:  $\alpha_{DFA}$  of daily log returns; grey bars: wild-bootstrap 95% band under RW3 (volatility kept, no autocorrelation)

## DFA Across Markets: Interpretation

- ▣ Inside the band: S&P 500 0.494, Euro Stoxx 50 0.488, Nikkei 0.503, PX 0.552, Shanghai 0.554
- ▣ Above the band (3 of 16): Ethereum 0.581, BET 0.580 (band up to 0.561), Bitcoin 0.576 (up to 0.575)
  - ▶ A shuffled band (upper limit about 0.55) flags 5 series: PX and Shanghai were “long memory” only through volatility clustering
- ▣ Cross-check with Lo's  $V$ : Bitcoin ( $V = 1.547$ ) is not rejected, BET and Ethereum marginally are
- ▣ Lesson: long memory in returns is weak and fragile; long memory in volatility (Chapter 1) is strong

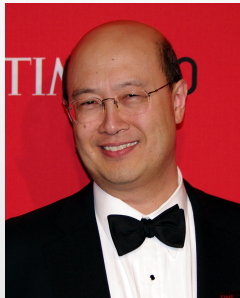
## Long Memory with Inference: the Local Whittle Estimator

- $d = H - 0.5$ ; spectral density  $f(\lambda) \sim G\lambda^{-2d}$  as  $\lambda \rightarrow 0$ ; fit only the first  $m$  Fourier frequencies  $\lambda_j = 2\pi j/T$ 
  - ▶  $\hat{d} = \arg \min_d \left[ \ln \left( \frac{1}{m} \sum_{j=1}^m \lambda_j^{2d} I(\lambda_j) \right) - \frac{2d}{m} \sum_{j=1}^m \ln \lambda_j \right]$ ,  $I$ : the periodogram
  - ▶ Robinson (1995): for  $|d| < 1/2$ ,  $m \rightarrow \infty$ ,  $m/T \rightarrow 0$ , a smooth  $f$  near 0 and a linear process with conditionally homoskedastic martingale-difference innovations,  $\sqrt{m}(\hat{d} - d) \rightarrow N(0, 1/4)$  without Gaussianity: a confidence interval instead of a band
- $m = T^{0.65}$ , 95% CI  $\hat{d} \pm 1.96/(2\sqrt{m})$ :
  - ▶ S&P 500  $-0.013$   $[-0.069, 0.043]$ ; Bitcoin  $0.063$   $[-0.002, 0.127]$
  - ▶ BET  $0.118$   $[0.062, 0.174]$ ; Ethereum  $0.112$   $[0.045, 0.179]$
- Bandwidth trade-off: a large  $m$  reaches frequencies where short-run autocorrelation (BET  $\hat{\rho}_1 = 0.109$ ) biases  $\hat{d}$  upward; with  $m = T^{0.5}$  BET still gives  $0.135$  (SE  $0.056$ ): a sensitivity check supporting, not proving, that its  $d > 0$  is not only short memory (GARCH returns also strain the conditions above)

## Recap: Long Memory

- ▣ The Hurst exponent measures how the range of cumulative returns scales with the horizon:  $H = 0.5$  means no long-range dependence (short memory allowed)
- ▣ The classical R/S Hurst slope is biased upward over small windows: always use the Anis–Lloyd correction
- ▣ Short memory mimics long memory: Lo's  $V$  separates them
- ▣ DFA needs a null that keeps volatility clustering (wild bootstrap); the local Whittle  $\hat{d}$  comes with a confidence interval
- ▣ Developed indices show no long memory in returns

## Andrew Lo and the Adaptive Market Hypothesis



Andrew W. Lo, MIT

Photo: David Shankbone (2012); CC BY 3.0;

Wikimedia Commons

- Lo (2004); book-length version Lo (2017)
- Markets are an ecology of competing species (traders, funds, algorithms)
  - ▶ Species adapt by trial and error, not by perfect optimisation
- Efficiency depends on the number of competitors, the profit opportunities and the environment
- Prediction: efficiency **varies over time**, and anomalies appear, weaken and may reappear

## EMH versus AMH

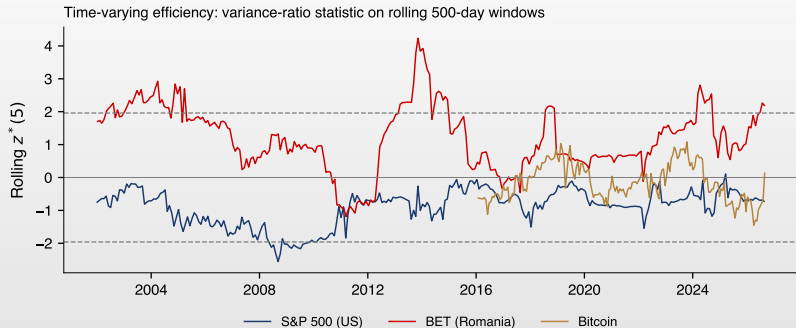
|                | EMH                                            | AMH                                                 |
|----------------|------------------------------------------------|-----------------------------------------------------|
| Investors      | Rational, optimising                           | Bounded rationality, heuristics                     |
| Efficiency     | A state: yes or no                             | A degree that changes with conditions               |
| Risk–return    | Equilibrium risk premia, possibly time-varying | Changes with the population of investors            |
| Anomalies      | Mispriced risk or data mining                  | Transient profit opportunities                      |
| Empirical tool | Tests of an equilibrium restriction            | Tests of time and state variation (rolling windows) |

- Evidence of time variation: survey [Lim & Brooks \(2011\)](#); long-run indices [Urquhart & McGroarty \(2016\)](#)

## Measuring Time-Varying Efficiency

- Compute a statistic on a **rolling window**: 500 trading days (about two years), moved forward by 21 days (one month)
  - ▶ Statistics:  $z^*(5)$  of the variance-ratio test and the DFA exponent
- Null band for DFA: wild bootstrap **inside each window** (199 sign flips), so the band follows the window's volatility
  - ▶ An i.i.d. Student- $t_4$  band  $[0.378, 0.614]$  ignores volatility clustering, which RW3 allows and which raises  $\alpha_{DFA}$
- Caveats
  - ▶ Overlapping windows are strongly correlated: runs of rejections are not independent evidence
  - ▶ Even under the null, about 5% of windows reject: compare the share of rejections with 5%

## Rolling Variance Ratio: S&P 500, BET, Bitcoin

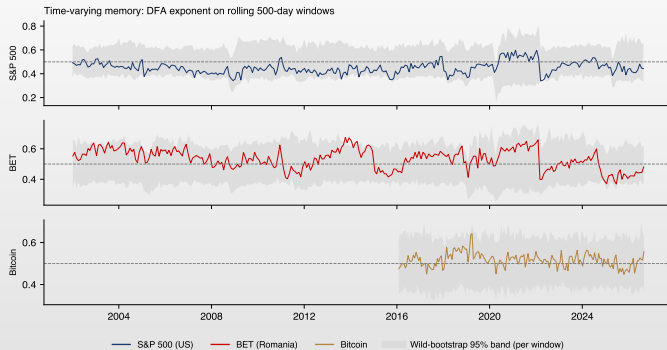


□  $z^*(5)$  on rolling 500-day windows; dashed lines  $\pm 1.96$

## Rolling Variance Ratio: Interpretation

- ▣ Share of windows with  $|z^*(5)| > 1.96$ : S&P 500 6.4%, BET 22.4%, Bitcoin 0.0%
- ▣ S&P 500: close to the 5% expected by chance; the full-sample rejection is concentrated in a few turbulent windows
- ▣ BET: rejections cluster in episodes, then fade: the pattern the AMH predicts
  - ▶ Compare the pre- and post-2020 periods in Seminar C1 (FTSE upgrade)
- ▣ Bitcoin, 2014–2026: no 500-day window rejects at 5%

## Rolling DFA Exponent: S&P 500, BET, Bitcoin



□  $\alpha_{DFA}$  on rolling 500-day windows; grey: wild-bootstrap 95% band computed in each window

## Rolling DFA: Interpretation

- Windows outside their own wild-bootstrap band: S&P 500 2.0%, BET 2.7%, Bitcoin 0.0%: all below the 5% expected by chance
- With the i.i.d. Student- $t_4$  band: 6.4%, 12.9%, 1.1%: the apparent BET “episodes” were volatility clustering
  - ▶ The wrong null over-rejects exactly where volatility is high: a test of RW1, not of RW3
- The band is wide (average [0.36, 0.64] for BET): 500 observations cannot pin down a scaling exponent
- Time variation in efficiency is better measured by the rolling  $z^*(5)$ , whose null is already robust

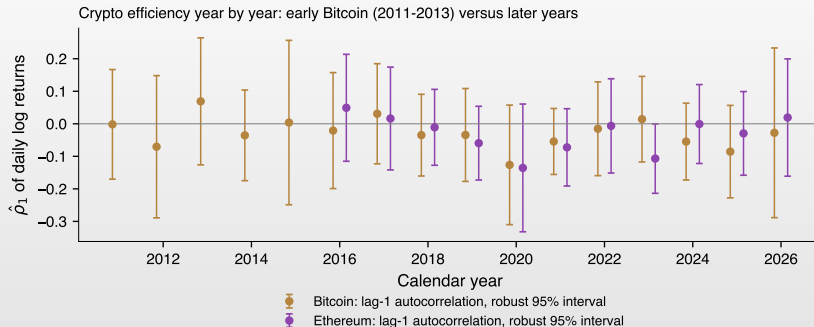
## The AMH, Tested

- Kim, Shamsuddin & Lim (2011): century-long US data; predictability varies with market conditions (crises, bubbles), not along a trend
  - ▶ This turns the AMH narrative into a regression with testable coefficients
- Our version: regress the rolling  $z^*(5)$  on the window's annualised volatility (%) and a crisis dummy (window overlaps Sep 2008–Jun 2009 or Feb–Jun 2020)
  - ▶ Newey–West errors with 24 lags: consecutive windows share 95% of their days
- S&P 500: crisis  $-0.67$  ( $t = -2.37$ ), volatility  $0.006$  ( $t = 0.27$ ),  $R^2 = 0.20$
- BET: crisis  $-0.82$  ( $t = -2.77$ ), volatility  $-0.008$  ( $t = -0.36$ ),  $R^2 = 0.15$
- Crisis windows are associated with lower  $z^*(5)$  (towards reversal) in both markets, beyond volatility: a conditional association, not a causal effect
  - ▶  $z^*$  mixes  $\widehat{\sqrt{R}} - 1$  and its standard error; state dependence is consistent with the AMH but does not identify it uniquely

## Is Bitcoin Efficient? The Literature

- [Urquhart \(2016\)](#): Bitcoin inefficient over 2010–2016; some tests indicate efficiency in the later subperiod
  - ▶ His battery includes BDS, which rejects under volatility clustering alone: not a test of RW3
- [Nadarajah & Chu \(2017\)](#): with a power transformation of returns, Bitcoin is weak-form efficient
- [Kristoufek \(2018\)](#): Bitcoin markets mostly inefficient in 2010–2017, with efficiency changing over time
- [Tran & Leirvik \(2020\)](#): time-varying efficiency across many crypto-assets, increasing over time
- [Pele et al. \(2023\)](#): crypto-assets have become a distinct asset class; their returns share the stylised facts of other markets

## Crypto Efficiency Year by Year



- Lag-1 autocorrelation of daily returns for each calendar year, with robust 95% intervals; Bitcoin from 2011

## Crypto Year by Year: Interpretation

- ▣ Only 1 of the 27 yearly  $\hat{\rho}_1$  has  $|t| > 1.96$  with robust errors: Ethereum 2023,  $\hat{\rho}_1 = -0.107$  (SE 0.054)
  - ▶ Chance alone would produce about  $0.05 \times 27 = 1.35$  such years
- ▣ Bitcoin 2011–2013:  $\hat{\rho}_1 = -0.002, -0.071, 0.069$ ; the largest  $|z^*(5)|$  in any year is 1.71
- ▣ This differs from [Urquhart \(2016\)](#), who rejects efficiency in early years
  - ▶ Different tests (Ljung–Box, runs, Bartels, automatic VR, BDS, R/S), a different price source and robust vs classical errors
  - ▶ A single year of 365 returns gives a robust SE of about 0.06–0.13: small effects are invisible
- ▣ The evidence against crypto efficiency depends strongly on the test and on the inference

## Romania: Frontier, then Emerging

- ▣ FTSE Russell (2019): Romania reclassified from Frontier to Secondary Emerging, effective 21 September 2020
- ▣ Expected channels: more foreign investors, more analysts, more liquidity  $\Rightarrow$  faster price discovery
- ▣ Earlier BVB evidence: crashes and log-periodic bubbles [Pele & Mazurencu-Marinescu \(2012\)](#); herding in an illiquid market [Pele, Mazurencu-Marinescu & Nijkamp \(2013\)](#)
- ▣ Did efficiency improve? Seminar C1 answers with a before/after test and a block bootstrap
  - ▶ Block bootstrap: resamples blocks of consecutive days to keep the dependence [Künsch \(1989\)](#)

## Recap: Adaptive Markets

- ▣ The AMH replaces “efficient or not” with “how efficient, when and why”
- ▣ Rolling tests reveal episodes: BET rejects in 22.4% of windows, the S&P 500 near the 5% chance level
- ▣ Rolling statistics need a null that keeps volatility clustering (wild bootstrap) and must be read as correlated evidence
- ▣ Crisis windows are associated with lower  $z^*(5)$  (towards reversal): state dependence consistent with the AMH, not a proof of it
- ▣ Crypto efficiency is test-dependent: 1 of 27 robust yearly tests rejects (Ethereum 2023), about what chance produces
- ▣ Market-structure events (FTSE upgrade) are natural experiments for efficiency

## The Event-Study Method

- ▣ Fama, Fisher, Jensen & Roll (1969): how fast do prices absorb public news (stock splits)?
- ▣ Market model on an estimation window before the event:  $r_{i,t} = a_i + b_i r_{m,t} + \varepsilon_{i,t}$ 
  - ▶  $r_{m,t}$ : market return;  $b_i$ : sensitivity of asset  $i$  to the market
- ▣ Abnormal return  $AR_{i,t} = r_{i,t} - \hat{a}_i - \hat{b}_i r_{m,t}$ ; cumulative  $CAR_i[\tau_1, \tau_2] = \sum_{t=\tau_1}^{\tau_2} AR_{i,t}$
- ▣ First-pass test:  $t = CAR_i / (\hat{\sigma}_\varepsilon \sqrt{L})$ , with  $L$  the length of the event window MacKinlay (1997); refinements two slides ahead
- ▣ Efficiency predicts a jump on day 0 and no drift or reversal afterwards

## Worked Example: the Romanian Bank Tax of December 2018

- ▣ Seminar 0, Exercise 7: tax announced on the evening of 18 Dec 2018; day 0 = 19 Dec; Banca Transilvania vs BET
- ▣ Market model on days  $[-250, -11]$ :  $\hat{b} = 1.18$ ,  $\hat{\sigma}_\varepsilon = 0.83\%$
- ▣ Day 0:  $AR_0 = -8.3\%$  ( $t = -10.0$ );  $CAR[0, +5] = -1.1\%$  ( $t = -0.54$ )
- ▣ Post-event window:  $CAR[+1, +5] = +7.2\%$ ,  $t = 3.82$  (full prediction-error covariance, next slide)
- ▣ Most of the day-0 fall reverses significantly within the week: consistent with an **overreaction**, against semi-strong efficiency
  - ▶ Or new information (the tax was later softened); a single event cannot separate the two: the joint-hypothesis problem again

## Event Studies: Inference Done Properly

- Prediction-error variance:  $\text{Var}(AR_\tau) = \sigma_\varepsilon^2 \left[ 1 + \frac{1}{T_0} + \frac{(r_{m,\tau} - \hat{\mu}_m)^2}{(T_0 - 1)\hat{\sigma}_m^2} \right]$ ,  $T_0$  = estimation days [MacKinlay \(1997\)](#)
  - ▶ Bank tax, BET  $r_{m,0} = -11.9\%$ : the bracket is 2.12, so TLV's  $t$  shrinks in absolute value from  $-10.0$  to  $-6.85$ ; BRD:  $AR_0 = -5.6\%$ ,  $t = -4.03$
- Several firms: standardise each  $AR$  by its own prediction error and use the cross-sectional variance, robust to event-induced variance [Boehmer, Masumeci & Poulsen \(1991\)](#)
- Same event day for all firms: abnormal returns are cross-correlated and cross-sectional tests over-reject [Kolari & Pynnönen \(2010\)](#)
  - ▶ Correction factor  $\sqrt{(1 - \bar{r}) / (1 + (N - 1)\bar{r})}$ ,  $\bar{r}$  = average residual correlation
- TLV and BRD:  $\bar{r} = 0.15$ ; mean  $AR_0$  with an independence  $t = -7.55$ , portfolio  $t = -7.03$ ; with ten such firms the variance of the mean  $AR$  is  $1 + 9\bar{r} = 2.38$  times its independence value, and a cross-sectional test, whose variance estimate is biased down by  $1 - \bar{r}$ , understates it 2.81 times

## How Fast Is “Fast”?

- Daily data can only show that news is absorbed within a day
- At high frequency, futures prices react to scheduled macro news within microseconds; [Tak, Schlamp, Osterrieder & Pele \(2026\)](#) measure this race at Eurex and CME
- Crypto events: [Pele, Lin, Wang & Gaman \(2025\)](#) study market responses to Ethereum development milestones
- Speed of adjustment is itself a measure of efficiency, and it depends on technology (Chapter 10, market microstructure)

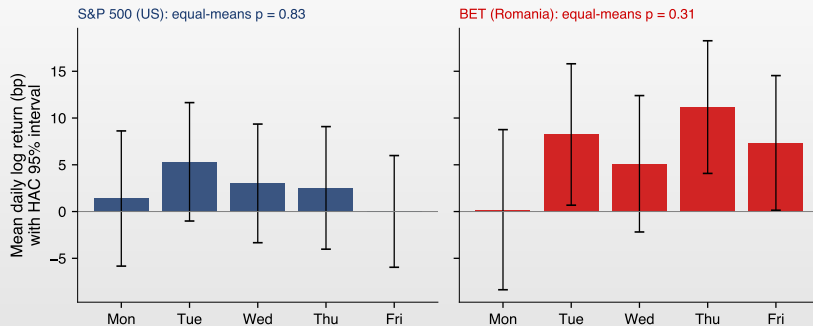
## Recap: Event Studies

- Semi-strong efficiency predicts a jump on day 0 and no drift or reversal afterwards
- On a large market move, the estimation error of the market model doubles the variance of  $AR_0$
- Clustered event days need tests that allow for cross-correlation (portfolio, Kolari–Pynnönen)
- The bank-tax fall survives every correction; the significant reversal over days 1–5 is consistent with overreaction, but one event cannot rule out later news

## Anomalies and Calendar Effects

- An **anomaly** is a return pattern the asset-pricing model cannot explain: mispriced risk, behavioural bias or data mining
  - ▶ Families: calendar, time-series (index momentum), cross-sectional (size, value, momentum: **Chapter 3**)
  - ▶ Returns 26% lower out of sample and 58% lower after publication [McLean & Pontiff \(2016\)](#)
- Calendar effects: Monday [French \(1980\)](#); January [Rozeff & Kinney \(1976\)](#); turn of the month (TOM: last trading day and first three of the next month) [Ariel \(1987\)](#); [Lakonishok & Smidt \(1988\)](#)
  - ▶ Day of week:  $r_t = \sum_{d=1}^5 \beta_d D_{d,t} + u_t$ , Wald test of  $\beta_1 = \dots = \beta_5$ ; TOM and January: one dummy each
  - ▶ Newey–West HAC errors [Newey & West \(1987\)](#),  $L = 5$  days or 3 months: residuals are heteroskedastic and autocorrelated

## Day-of-the-Week Effect: S&P 500 and BET



□ Mean daily log return by weekday (bp), 2000–2026, with HAC 95% intervals

## Day of the Week: Interpretation

- ▣ S&P 500: means between 0.0 bp (Friday) and 5.3 bp (Tuesday); equal-means  $p = 0.83$
- ▣ No Monday effect since 2000: the anomaly of French (1980) has disappeared
- ▣ BET: Thursday 11.2 bp ( $t = 3.09$ ), Monday 0.2 bp; yet the joint test gives  $p = 0.31$
- ▣ A single large  $t$  among five days is expected by chance: test the joint hypothesis
  - ▶ Individual  $t$  tests against zero also mix the “day effect” with the positive average return

## January and Turn-of-the-Month Effects

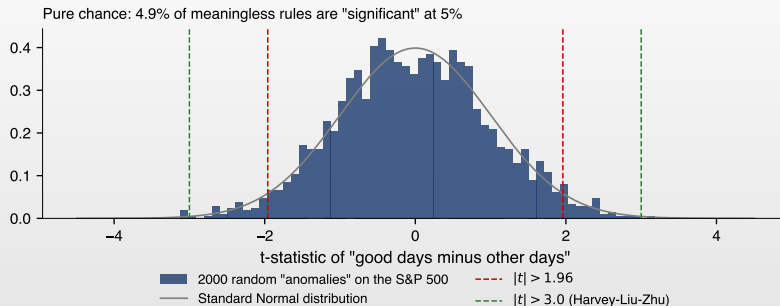
| Market         | January extra (%/month) | $t_{HAC}$ | TOM extra (bp/day) | $t_{HAC}$ | $p$    |
|----------------|-------------------------|-----------|--------------------|-----------|--------|
| S&P 500 (US)   | -0.66                   | -0.80     | 3.2                | 0.93      | 0.3539 |
| BET (Romania)  | 3.48                    | 1.49      | 17.3               | 3.62      | 0.0003 |
| Euro Stoxx 50  | -0.30                   | -0.29     | 5.1                | 1.21      | 0.2266 |
| WIG20 (Poland) | 0.53                    | 0.40      | 8.2                | 1.73      | 0.0841 |
| BUX (Hungary)  | 2.66                    | 2.25      | 15.1               | 3.63      | 0.0003 |

- TOM: BET +17.3 bp per day and BUX +15.1 bp, both with  $|t| > 3.5$ ; S&P 500 only +3.2 bp
- January: large but imprecise for BET ( $t = 1.49$ ); BUX, with  $t = 2.25$ , is the only market with a January-effect  $p$ -value below 0.05
- We ran 15 tests (5 markets  $\times$  3 effects): which survive a correction for multiple testing? Next slides

## Multiple Testing: Bonferroni and Holm

- With  $M$  independent tests at level  $\alpha$ ,  $\text{FWER} = 1 - (1 - \alpha)^M$ 
  - ▶  $M = 15$ ,  $\alpha = 5\%$ :  $\text{FWER} = 53.7\%$ , a coin flip
- **Bonferroni**: reject only if  $p_i < \alpha/M$ ; guarantees  $\text{FWER} \leq \alpha$  for any dependence
- **Holm**: sort  $p_{(1)} \leq \dots \leq p_{(M)}$ ; reject while  $p_{(j)} < \alpha/(M - j + 1)$ , stop at the first failure
  - ▶ Same guarantee, never less powerful than Bonferroni
- Our 15 calendar tests: Bonferroni threshold 0.0033; 3 raw rejections, 2 survive Holm (BET and BUX turn of the month)
- Seminar B6 carries out the full correction

## How Many “Anomalies” Does Pure Chance Produce?



- 2000 meaningless rules on the S&P 500: each labels a random 20% of days as “good” and tests good minus other days

## Pure Chance: Interpretation

- 4.9% of the random rules have  $|t| > 1.96$ , exactly the 5% the test promises; the largest  $|t|$  is 3.19
- A researcher who tries many rules and reports the best one will always find a “significant anomaly”
  - ▶ Data snooping in technical trading rules: [Sullivan, Timmermann & White \(1999\)](#)
- [Harvey, Liu & Zhu \(2016\)](#): after hundreds of published factors, a new one should have  $|t| > 3.0$ ; here 0.25% of random rules pass
- BET and BUX turn of the month ( $|t| > 3.5$ ) survive even this stricter threshold

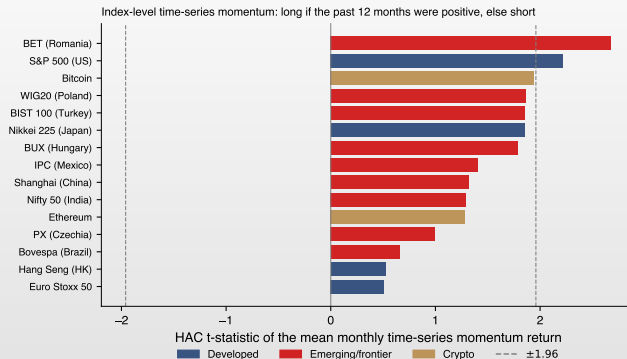
## Dependent Tests: Reality Check, SPA and StepM

- Bonferroni and Holm ignore the correlation between tests (same markets, same days)
  - ▶ [White \(2000\)](#): Reality Check, bootstrap of  $\max_j$  over all rules tried; [Hansen \(2005\)](#): SPA, studentised and less sensitive to poor rules
  - ▶ [Romano & Wolf \(2005\)](#): StepM, step-down control of the FWER from the **joint** bootstrap distribution of all statistics
- Our 15 calendar tests: StepM with a bootstrap of blocks of 3 calendar months, common to all markets, 999 replications
  - ▶ Critical  $p$  at the first step 0.0016 (Bonferroni: 0.0033); 2 rejections, BET and BUX TOM, as with Holm
  - ▶ Why stricter than Bonferroni? At nominal 5%, the bootstrap rejection rates of the HAC tests are 6.0% (day of week), 6.4% (TOM), 6.4% (January)
- From over two million simulated strategies: thresholds  $|t| > 3.8$  (time series) and 3.4 (cross-section); ignoring them makes about 45% of rejections false [Chordia, Goyal & Saretto \(2020\)](#); lowering false discoveries raises missed ones [Harvey & Liu \(2020\)](#)

## Momentum and Reversal

- **Reversal:** past long-term losers (3–5 years) outperform past winners [De Bondt & Thaler \(1985\)](#)
- **Cross-sectional momentum:** stocks that rose most over 3–12 months keep outperforming [Jegadeesh & Titman \(1993\)](#) (Chapter 3)
- **Time-series momentum (TSMOM)** of a single index [Moskowitz, Ooi & Pedersen \(2012\)](#):
  - ▶ Signal  $s_t = \text{sign}\left(\sum_{j=1}^{12} r_{t-j}\right)$  from the past 12 monthly log returns
  - ▶ Strategy return  $s_t R_t$ ,  $R_t$  the simple monthly return: long after a positive year, short after a negative one, before financing and costs
  - ▶ [Moskowitz, Ooi & Pedersen \(2012\)](#) use futures excess returns scaled to 40% ex-ante volatility; our price indices lack financing rates and dividends: a data limitation
  - ▶ Test: HAC  $t$  of the mean of  $s_t R_t$  (6 lags); Sharpe ratio  $\sqrt{12} \bar{x} / s_x$

## Index-Level Time-Series Momentum



□ HAC  $t$ -statistic of the mean monthly TSMOM return, 2000–2026 (crypto from 2014/2016)

## Time-Series Momentum: Interpretation

- ▣ Significant at 5%: BET ( $t = 2.68$ , 1.40% per month) and S&P 500 ( $t = 2.22$ )
- ▣ But the Sharpe ratio stays below buy-and-hold: BET 0.65 vs 0.75, S&P 500 0.48 vs 0.52
  - ▶ Much of the TSMOM return is the equity premium earned while long
- ▣ BET's monthly autocorrelation is 0.19: the same persistence found in daily data
- ▣ Bitcoin:  $t = 1.94$ ; the sample contains only 307 months for equities and far fewer for crypto, limiting test power
- ▣ 15 markets tested: the multiple-testing caution applies here too

## Recap: Anomalies

- ▣ Test calendar effects with regressions on dummies and HAC standard errors; test all days jointly
- ▣ The Monday effect is gone; the turn of the month survives in BET and BUX even after Holm correction
- ▣ Pure chance produces 5% “anomalies”: correct for the number of tests (Bonferroni, Holm,  $|t| > 3$ )
- ▣ Index momentum is statistically visible in BET and S&P 500 but does not beat buy-and-hold on a Sharpe basis
- ▣ Cross-sectional anomalies and the factor zoo: Chapter 3

## Robert Shiller and Richard Thaler



Robert J. Shiller, Nobel Prize 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

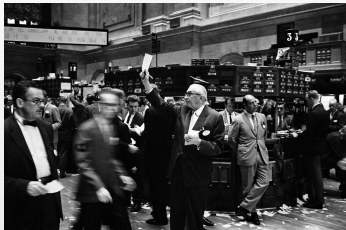


Richard H. Thaler, Nobel Prize 2017

Photo: Bengt Nyman (2017); CC BY 2.0; Wikimedia Commons

- Shiller (1981): stock prices move far more than the dividends that should justify them (excess volatility)
- De Bondt & Thaler (1985): investors overreact; extreme losers become winners over the next 3–5 years

## Psychology in Prices, from the Floor to the Feed



New York Stock Exchange trading floor, 1963

Photo: Thomas J. O'Halloran (1963); public domain;  
Wikimedia Commons

- **Prospect theory** [Kahneman & Tversky \(1979\)](#): losses loom larger than gains relative to a reference point
  - ▶ Disposition effect: winners sold too early, losers held too long
- **Overconfidence**: too much trading; **anchoring** and **limited attention**: slow reaction to news, a candidate cause of momentum
- **Herding** [Pele, Mazurencu-Marinescu & Nijkamp \(2013\)](#): once voices on a floor, now order books, algorithms and social media (Chapter 10)
- Biases alone do not break efficiency: rational arbitrageurs could trade against them. Can they?

## Limits to Arbitrage



Andrei Shleifer, Harvard

Photo: Manuel Castells (2018); CC BY-SA 2.0;

[Wikimedia Commons](#)

- **Shleifer & Vishny (1997)**: real arbitrage is risky and needs capital
- **Noise-trader risk**: mispricing can get worse before it corrects
  - ▶ Noise traders: investors trading on non-information (sentiment, rumours)
- Fund managers lose clients after losses and must close positions at the worst time
- Costs: short-selling limits, margins, trading costs, no perfect substitutes
- Hence anomalies can persist where arbitrage is hard: small, illiquid, frontier markets

## Recap: Behavioural Finance

- ▣ Excess volatility and overreaction challenge the EMH; prospect theory and herding offer mechanisms
- ▣ Biases matter for prices only when arbitrage is limited by risk, capital and costs
- ▣ Where arbitrage is hardest (small, illiquid markets like BET) we indeed find the strongest persistence
- ▣ The AMH reconciles both views: rational and biased behaviour coexist and evolve

## Can Machine Learning Beat Efficient Markets?

- ▣ ML models capture nonlinear patterns that autocorrelation tests miss
  - ▶ ML: machine learning, algorithms that learn a prediction rule from data (Chapter 13)
- ▣ [Gu, Kelly & Xiu \(2020\)](#): trees and neural networks give large economic gains over linear regressions in the US cross-section
- ▣ Yet their monthly out-of-sample  $R^2$  is below 1%: consistent with an almost efficient market
- ▣ Large language models read news and filings: [Tak & Pele \(2025\)](#) forecast stock trends with LLaMA
- ▣ AMH reading: every successful model attracts competitors, and its edge decays

## ML and Efficiency: Pitfalls

- ▣ **Data snooping:** many models on one dataset produce a “winner” by chance (as in our random-rules experiment)
- ▣ **Look-ahead bias:** using information not available at the time of the forecast
- ▣ **Costs:** high-turnover strategies lose their edge after trading costs
- ▣ **Non-stationarity:** a pattern found in 2000–2015 may be gone by 2026 ([McLean & Pontiff \(2016\)](#))
- ▣ Chapter 13 shows walk-forward validation and the deflated Sharpe ratio to guard against these

## Big Data and Efficiency: Research Question

- [Martin & Nagel \(2022\)](#): investors price  $N$  assets whose cash flows depend on  $J$  characteristics with unknown coefficients; what does a rejected no-predictability test mean when  $J$  is comparable to  $N$ ?
  - ▶ the coefficients must be learned from past data: a high-dimensional prediction problem, as in machine learning
- Semi-strong efficiency: prices “fully reflect” public information [Fama \(1970\)](#); two readings
  - ▶ investors know the parameters of the cash-flow model (rational expectations: the null of most efficiency tests)
  - ▶ investors update their beliefs optimally (Bayes) with the data available in real time
- Why it matters: hundreds of in-sample return predictors, the “factor zoo” [Harvey, Liu & Zhu \(2016\)](#); text, satellite images and social media keep increasing  $J$ 
  - ▶ does every significant predictor need a risk-based or a behavioural explanation?
- Link to this chapter: in-sample vs out-of-sample predictability [Campbell & Thompson \(2008\)](#), [Goyal, Welch & Zafirov \(2024\)](#), and the decay of anomalies after publication [McLean & Pontiff \(2016\)](#)
- Investors are risk-neutral in the model: no joint-hypothesis problem, so no predictability can be a risk premium

## Big Data and Efficiency: Model and Method

- **Model** (Sec. II): cash-flow growth  $\Delta y_t = Xg + e_t$ ,  $X$  the  $N \times J$  matrix of characteristics,  $e_t \sim N(0, I)$ ; zero interest rate
  - ▶ prior  $g \sim N(0, (\theta/J)I)$ : the posterior mean is a ridge regression (OLS with a penalty on the squared coefficients), i.e. coefficients shrunk towards 0 (Prop. 1)
  - ▶ realised return = underreaction caused by shrinkage + estimation noise + unpredictable shock (Prop. 2)
- **Tests**: in sample, a cross-sectional OLS of  $r_{t+1}$  on  $X$  with a joint  $\chi^2$  test; out of sample, a portfolio with weights  $\frac{1}{N}Xh_{s+1}$  from coefficients estimated in other periods
  - ▶  $r_{\text{OOS},t+1} = \frac{1}{N}h'_{s+1}X'r_{t+1}$ ,  $s \neq t$  (eq. 19)
- **Asymptotics and simulations**:  $N, J \rightarrow \infty$  with  $J/N \rightarrow \psi > 0$ ; simulations with  $N = 1000$ , standard-Normal entries of  $X$  and  $\theta = 1$  (Sec. IV)
- **Empirical design** (Sec. VII): US stocks in CRSP, without those below the 20th NYSE size percentile or priced below \$1
  - ▶ predictors: simple and squared monthly returns at lags  $t - 2$  to  $t - 120$  (238 predictors), demeaned and standardised across stocks every month
  - ▶ full-sample ridge, 1971 – June 2019, penalty chosen by leave-one-year-out cross-validation; rolling 20-year OLS gives  $r_{\text{OOS}}$  from January 1956
  - ▶ figure numbers refer to the [working-paper version](#) (May 2021)

## Big Data and Efficiency: Key Results

- In sample: the rejection probability of the no-predictability test tends to 1 as  $N, J \rightarrow \infty$  (Prop. 4); for  $\psi > 0.4$  ( $\theta = 1$ ,  $t = 1$ ) the probability of no rejection is of the order  $e^{-0.015N}$  (Fig. II)
  - ▶ with fully rational Bayesian investors, the rational-expectations null is almost surely rejected: a rejection is not evidence of irrationality
- Simulations ( $N = 1000$ , four periods of learning): a test of nominal size 5% rejects with probability above 90% once  $J > N/2$  (Fig. IIIb)
  - ▶ not a size distortion: the in-sample predictable component really is in the prices
- Single-predictor tests with  $k$ -FWER control: for  $k = 20$  and  $J/N = 0.05$ , 25% rejections instead of 5%; about 80% for  $J/N$  close to 1 (Fig. IV)
  - ▶ multiple-testing corrections do not help: the problem is the null hypothesis, not the test
- Out of sample:  $E[r_{\text{OOS},t+1}] = 0$ , forward and backward in time (Prop. 6); positive only if investors shrink more than the data warrant (Fig. VI)
  - ▶ out-of-sample tests keep their economic meaning
- US stocks: the ridge coefficients recover momentum, long-run reversal and momentum seasonality (Fig. VII);  $r_{\text{IS}} \approx 0.006\%$  per month, an in-sample  $R^2$  of about 1%, stable over time (Fig. VIII)
  - ▶  $r_{\text{OOS}}$  is much lower and often near 0; in the 10-year windows ending in 2014–2019 it is less than two standard errors above 0: little of the in-sample predictability survives

## Big Data and Efficiency: Replicate, Limits, Extend

- **Replicate with the course data** (stock-level CRSP files are not part of them)
  - ▶ 49 industry portfolios, monthly value-weighted returns from July 1926 ([Kenneth French Data Library](#)); the same 238 predictors, monthly standardisation and rolling 20-year windows (Seminar 2, C3)
  - ▶ data limitation: 49 portfolios instead of thousands of stocks; the size and price filters do not apply to portfolios
- **Limits stated by the authors**
  - ▶ lagged returns are equilibrium outcomes, not the exogenous  $X$  of the model: the application illustrates the wedge, it does not test the model
  - ▶ a  $t$ -test on the monthly  $r_{\text{OOS}}$  is not a valid test of  $E[r_{\text{OOS}}] = 0$ : the expectation also averages over  $g$
  - ▶ the backward result may rely on i.i.d. cash-flow growth; homogeneous investors and constant characteristics
- **Open extensions**
  - ▶ time series, as the authors suggest: does the dividend–price regression of this chapter show the same in-sample/out-of-sample wedge?
  - ▶ nonlinear predictors (trees, neural networks [Gu, Kelly & Xiu \(2020\)](#)) enlarge  $J$ : does the wedge grow?
  - ▶ AMH: is early out-of-sample predictability the excess shrinkage of investors with less computing power?

## An Open Question: Did Spot ETFs Change Bitcoin's Efficiency?

- US spot Bitcoin ETFs trade from 11 January 2024: new arbitrageurs, new liquidity, a new venue for price discovery
- [Kia, Liu, Li, Song & Xu \(2025\)](#): with a fractionally cointegrated VAR on minute data, the ETF led the spot market in price discovery in 2021–2022
  - ▶ after the spot ETFs, the spot market became the dominant venue again
- [Tran & Leirvik \(2020\)](#): crypto efficiency varies over time and has increased: the AMH view of this chapter
- Open: does a change in market structure change weak-form efficiency, and can a before/after design identify it?
  - ▶ confounders in 2024: the April halving, the rate cycle, record inflows

## The Discovery Loop, Step by Step (1/2)

### □ Literature

- ▶ List peer-reviewed studies since 2020 on how ETF introductions change the efficiency or price discovery of the underlying. For each: DOI, design (event study, DiD, structural break), efficiency measure. Flag any entry you are unsure exists.
- ▶ Resolve every DOI on Crossref; the title and authors must match

### □ Hypothesis

- ▶ Give three mechanisms by which spot ETFs could change Bitcoin's serial dependence (arbitrage, investor mix, trading hours). For each, the sign of the change in  $\rho_1$  and  $VR(q)$ , and a variable that separates it from the others.

### □ Data and code

- ▶ Read BTC-USD.CC from data/market (close); log returns;  $\rho_1$  with a heteroskedasticity-robust standard error and the Lo-MacKinlay  $VR(5)$  with  $z^*$ , in two windows of equal length around 11 January 2024.
- ▶ Replicate a known number first: the chapter's full-sample Bitcoin  $\hat{\rho}_1 = -0.022$  (robust SE 0.026)

## The Discovery Loop, Step by Step (2/2)

### ▣ Robustness

- ▶ Write a pre-registered robustness plan: window lengths, weekday-only closes,  $VR(q)$  for  $q = 2$  to 20 with Chow-Denning, placebo dates, and Ethereum as a control until its own spot ETFs (July 2024).

### ▣ Critique

- ▶ Act as a hostile referee: if Bitcoin was already efficient before 2024, what could this test ever detect? Compute the minimum detectable change in  $\rho_{01}$  and name a better outcome variable.

### ▣ Report

- ▶ Estimates, robust standard errors, the placebo distribution, every test run (with a multiple-testing correction)
- ▶ Prompts that shaped a decision go into `AI_USE.md`

- ▣ The assistant drafts the tests; the human decides which one answers the question

## What the Researcher Must Check

- **i.i.d. standard errors:**  $1/\sqrt{T}$  rejects too often under volatility clustering
  - ▶ test: robust SE and  $z^*$ , as in the weak-form tests of this chapter
- **Unequal windows:** different power before and after; test: equal lengths, fixed in advance
- **Event date chosen from the data**
  - ▶ test: the regulatory date only, plus a placebo distribution at dates without an event
- **Confounding:** before/after is not causal; test: a control asset (Ethereum before July 2024) in a difference-in-differences
  - ▶ causal only with parallel pre-trends, no anticipation and no spillover from Bitcoin ETFs to Ethereum; the common post-period (Jan–Jul 2024) is short; otherwise the comparison is descriptive
- **Many statistics:**  $\rho_1$ , VR(2–20), runs; test: Chow–Denning or Holm, not the best single  $p$ -value
- **Low power:** report the minimum detectable change,  $\approx 2.8$  standard errors of the difference

## Mini-Case: Did Bitcoin Become More Efficient?

- A typical assistant proposal: “spot ETFs brought arbitrageurs, so Bitcoin became more efficient”
- Test: two windows of 982 days around 11 January 2024;  $H_0 : \rho_{1,\text{post}} = \rho_{1,\text{pre}}$ 
  - ▶  $z = \Delta\hat{\rho}_1 / \sqrt{s_{\text{pre}}^2 + s_{\text{post}}^2}$ , robust standard errors  $s$
- Before (May 2021 – Jan 2024):  $\hat{\rho}_1 = -0.036$  ( $s = 0.042$ ),  $\text{VR}(5) = 1.000$  ( $z^* = 0.00$ )
- After (Jan 2024 – Sep 2026):  $\hat{\rho}_1 = -0.050$  ( $s = 0.049$ ),  $\text{VR}(5) = 0.939$  ( $z^* = -0.64$ );  $\Delta\hat{\rho}_1 = -0.014$ ,  $z = -0.21$
- Placebo at 11 January 2016–2021: 0 of 6 significant, largest  $|z| = 1.57$
- Interpretation: neither window rejects the null; these tests do not establish whether smaller inefficiencies existed before the ETFs
  - ▶ minimum detectable change in  $\rho_1$ : 0.18; the question needs another outcome, such as intraday price discovery

 ai\_discovery\_case.py

## Your Turn: A Project Seed

- **Question:** did the spot ETFs change the time-varying efficiency of Bitcoin, relative to a control asset?
  - ▶ rolling VR and DFA (this chapter) for Bitcoin and Ethereum; Ethereum is untreated until July 2024
  - ▶ fix a common window ending July 2024 and discuss parallel trends, anticipation and spillovers; without them, report descriptive, not causal, differences
- **Replicate first:** the chapter's yearly Bitcoin autocorrelations and the mini-case numbers
- **Deliverables**
  - ▶ a pre-registration note ( $H_0$ ,  $H_1$ , dates, windows, corrections) committed before the results
  - ▶ code as a Quantlet; a report with robust inference, placebo tests and power
- **Logs:** prompts in `AI_USE.md`; every caught error (an i.i.d. standard error, an invented reference) in `AI_ERRORS.md`

## Conclusions

- ▣ Efficiency means unpredictable excess returns given the information set; weak-form tests are the cleanest
- ▣ Robust inference is essential: volatility clustering makes classical tests reject too often
- ▣ 2000–2026: S&P 500 shows mild reversal, BET and EUR/RON persistence, Bitcoin no robust rejection by autocorrelation and VR tests (its DFA exponent is above the band)
- ▣ Long memory in returns is weak; apparent memory is often short memory or volatility clustering
- ▣ The dividend–price slope is insignificant in sample even before the Stambaugh correction, and out of sample it does not beat the historical mean
- ▣ Efficiency varies over time (AMH); anomalies must survive multiple-testing corrections and costs
- ▣ Statistical rejection  $\neq$  exploitable profit  $\neq$  irrational market

## Further Study on Quantinar

### Advanced courses on quantinar.com

-  Efficiency of cryptocurrency markets - A GMM-based analysis
-  Generalized Method of Moments

### Code, data and textbooks

- All charts and tables: [github.com/danpele/MFM/tree/main/Quantlets/Ch\\_02](https://github.com/danpele/MFM/tree/main/Quantlets/Ch_02)
- Textbooks: Campbell, Lo & MacKinlay (1997), Ch. 2; Franke, Härdle & Hafner (2019)

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